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Predicting Residential Building Energy Performance: A Comparison of OLS, Lasso, Eco-RETINA, Random Forest and Neural Networks

Abstract

This paper compares five predictive modelling approaches applied to the estimation of residential building heating load (kWh/m²): Ordinary Least Squares (OLS), Lasso, Eco-RETINA, Random Forest (RF) and a Multilayer Perceptron Neural Network (NN). The dataset is drawn from Tsanas y Xifara (2012), comprising 768 simulated residential buildings described by geometric and glazing characteristics. All models are estimated on the same training set (N=576) and evaluated on the same holdout test set (N=192), ensuring full comparability out-of-sample.

The econometric interest of this exercise lies in the structural multicollinearity present in the dataset: since all buildings share the same volume, geometric variables such as relative compactness and surface area are analytically related ($r=-0.992$), as are roof area and overall height ($r=-0.973$). This is not a statistical artefact but a physical consequence of the experimental design, making the dataset a demanding benchmark for linear models. A similar challenge arises in real-world settings: Raushan et al. (2024) document that 30% of entries in Ireland's national Energy Performance Certificate database contain errors, predominantly in geometric variables.

Data and Methodology

The dependent variable is heating load, measured in kWh/m². The six continuous regressors are *relative compactness, surface area, wall area, roof area, overall height and glazing area*. Two categorical variables: *building orientation* (4 categories) and *glazing area distribution* (6 categories); are encoded as dummy variables, with one reference category omitted per variable to avoid perfect multicollinearity (orientation_2 and glazing_dist_1). The final model includes 14 regressors plus a constant. The 75%/25% train-test split is fixed across all models.

OLS and Eco-RETINA are estimated with HC3 heteroskedasticity-robust standard errors, whose presence is confirmed by the Breusch-Pagan test ($LM=271.66$, $p<0.001$). Residual non-normality (Jarque-Bera=25.02, $p<0.001$) is a structural feature arising from the discrete nature of overall height (two values: 3.5m and 7.0m), which generates two distinct clusters of observations. Lasso is applied to standardised inputs with the penalty parameter λ selected by 10-fold cross-validation ($\lambda=0.009$). RF is estimated with 500 trees, max_features=$\sqrt{p}$ and no depth constraint. The Neural Network uses a relu(64,32) architecture; results reported as the mean over seven random seeds for robustness.

Results: out-of-sample predictive metrics for all five models

Model	R²_test	RMSE_test	MAE_test	MAPE_test
OLS	0.911	3.025	2.184	10.23%
Lasso	0.911	3.026	2.198	10.31%
Eco-Retina	0.923	2.820	2.088	10.14%
Random Forest	0.987	1.175	0.823	4.99%
Neural Network	0.993*	0.857*	0.588*	3.50%*

*Table 1. Out-of-sample metrics for all five models (N=192). *Mean over 7 random seeds*

OLS achieves a reasonable fit (R^2_test=0.911, RMSE_test=3.025 kWh/m²) but exhibits two structural limitations: severe multicollinearity among geometric variables (VIF diverges for surface area, wall area and roof area) and a systematic compression-toward-the-mean pattern in predictions, indicating that the underlying relationship is not purely additive. Empirical coverage of 95% prediction intervals is 85.9%, below the nominal level due to residual non-normality.

Lasso produces virtually identical results to OLS (RMSE_test=3.026), retaining 12 of 14 regressors. This confirms that L1 regularisation does not resolve multicollinearity of physical origin: shrinking coefficients cannot eliminate a dependence imposed by the constant-volume constraint.

Eco-RETINA improves moderately on the linear benchmarks (R^2_test=0.923, RMSE_test=2.820 kWh/m²). Its main contribution is interpretive as well as predictive: the algorithm automatically selects interaction terms that linear models cannot capture. The most influential term is overall_height × glazing_area ($z=39.60$, $p<0.001$), indicating that the effect of glazing on heating load is amplified as building height increases. The product relative_compactness × surface_area ($z=10.02$, $p<0.001$) resolves endogenously the multicollinearity that destabilises OLS estimates. Orientation variables remain non-significant across all specifications ($p>0.88$), consistent with Tsanas and Xifara (2012).

Random Forest achieves a substantial improvement (R^2_test=0.987, RMSE_test=1.175 kWh/m², MAPE=4.99%). The gap between train (R^2=0.998) and test is modest, indicating controlled overfitting. Permutation-based variable importance confirms glazing_area_modified as the dominant predictor (R^2 drop=0.131), with overall height, relative compactness, surface area and roof area contributing approximately equally (~0.10–0.12). Orientation variables show zero or negative permutation importance across all seeds. RF does not produce individual coefficients or formal prediction intervals.

The Neural Network obtains the best predictive performance (R^2_test=0.993, RMSE_test=0.857 kWh/m², MAPE=3.50%, averaged over seven seeds). The mean difference between train R^2 (0.9995) and test R^2 (0.9926) is 0.007 points, ruling out substantial overfitting. Convergence stability required removing early stopping (max_iter=5000, tol=1e-5); with this configuration, six of seven seeds produce RMSE_test below 1.0 kWh/m². As with RF, the NN produces no individual coefficients or prediction intervals.

Conclusions

The results confirm that the relationship between building characteristics and heating load is non-linear and interaction-driven. The structural multicollinearity of the dataset poses a fundamental challenge for linear models that neither OLS nor Lasso can overcome. Eco-RETINA partially addresses this by selecting interaction terms automatically while preserving full econometric interpretability — individual coefficients, hypothesis tests and prediction intervals. RF and the Neural Network resolve it implicitly through their non-parametric structure, at the cost of reduced transparency.

The choice of model should be governed by the analyst's objective. If predictive accuracy is paramount, the Neural Network is the superior alternative. If interpretability is required alongside high precision, Random Forest offers a solid compromise through variable importance measures. If the goal is to retain full econometric structure, Eco-RETINA is the preferred option. OLS and Lasso remain useful as linear benchmarks but are insufficient to capture the full structure of the problem.

A key limitation is that the data originate from building energy simulations rather than observed consumption records. Future work could extend the comparison to real-world EPC databases, incorporate the cooling load as a second dependent variable, or assess model stability through repeated cross-validation.

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